

Measure of Central tendency

Online Store

They use **Mean, Median, Mode, and Standard Deviation together** to manage inventory and avoid **stockouts or overstock**.

Real Inventory Forecasting Example (50 Days Data) Imagine an **online store selling wireless keyboards**. The system records **daily sales for 50 days**.

Dataset (Units Sold Per Day)(50 days sales)

[18,20,22,19,21,23,24,22,20,21, 25,26,24,23,22,21,20,19,18,22,
23,24,25,26,27,24,23,22,21,20, 19,18,20,22,23,24,26,27,25,24,
23,22,21,20,19,18,22,23,24,25]

Mean (Average Demand) Mean tells the company:

“On average, how many products sell per day?”

Suppose total units sold in 50 days = **1120**

Mean:

$$Mean = \frac{1120}{50} = 22.4$$

```
import numpy as np

sales = [15,18,20,22,19,21,23,20,18,25,
         24,26,20,22,21,19,23,24,22,20,
         18,21,23,25,26,24,22,20,21,23,
         24,26,27,25,23,22,21,20,19,18,
         22,23,24,26,27,25,24,23,22,21]

print("Mean:", np.mean(sales))
```

✓ 0.6s

Mean: 22.14

Interpretation

The store sells **about 22–23 units per day**.

Inventory Use

Store manager can stock:

23 units/day × 7 days = 161 units per week

Decision

Warehouse must stock at least:

$22.6 \times 30 \text{ days} \approx 678 \text{ units per month}$

So purchasing department orders **~700 keyboards per month**.

Median (Typical Demand)

Median = **middle value of sorted data**

Median tells the **typical demand** without being affected by unusual days.

For **50 observations**

$$\text{Median} = \frac{25\text{th} + 26\text{th}}{2}$$

Suppose:

25th = **22**

26th = **23**

$$\text{Median} = 22.5$$

```
import numpy as np

sales = [15,18,20,22,19,21,23,20,18,25,
         24,26,20,22,21,19,23,24,22,20,
         18,21,23,25,26,24,22,20,21,23,
         24,26,27,25,23,22,21,20,19,18,
         22,23,24,26,27,25,24,23,22,21]

print("Median:", np.median(sales))
```

✓ 0.0s

Median: 22.0

Why Median Matters

Suppose a **Black Friday sale** happens:

Normal day sales = 22

Sale day = 200

Mean becomes very large.

But **median still shows normal demand**.

Companies like Amazon rely heavily on **median demand** for baseline forecasting.

Mode (Most Frequent Sales)

Mode tells us **most common demand**.

Example:

Mode = 22

Meaning:

Most days **22 units are sold**.

Inventory Decision

Warehouse may use **mode as base stock level**.

Example:

Base Stock = 22 units/day

Standard Deviation (Demand Uncertainty)

Standard Deviation (Demand Variation)

Standard deviation tells us **how much sales fluctuate**.

Example result:

SD = **2.8**

Interpretation

Most sales fall between:

$$22.4 \pm 2.8$$

Range:

19 → 25 units

Inventory Use

Helps calculate **Safety Stock**.

Why Standard Deviation is Critical

Inventory managers calculate **Safety Stock**.

```
import numpy as np

sales = [15,18,20,22,19,21,23,20,18,25,
         24,26,20,22,21,19,23,24,22,20,
         18,21,23,25,26,24,22,20,21,23,
         24,26,27,25,23,22,21,20,19,18,
         22,23,24,26,27,25,24,23,22,21]

print("Standard Deviation:", np.std(sales))
```

✓ 0.0s

Standard Deviation: 2.6458269028793246

How AI Uses These Statistics

In AI / Machine Learning demand forecasting:

Algorithms use:

- Mean → average demand
- Median → robust central value
- Mode → most frequent demand
- SD → demand variability